Variable Acceleration

Functions of Time

1. A particle has velocity $v = \frac{1}{2}t^3$ m/s. Find its velocity after 4 s, and the time when $v = 108$ m/s.
2. Another particle has $v = 2t^2 - 16t + 24$, for $t \geq 0$. Find:

a. initial velocity;

b. times at rest;

c. time when $v = 64$;

d. greatest speed on $0 \leq t \leq 5$.

Solution 2

1

1. $v(4) = 32$ m/s.

   $108 = \frac{1}{2}t^3 \Rightarrow t = 6$ s.

2

2. a. $v(0) = 24$ m/s.

b. $2(t-2)(t-6) = 0$: $t = 2, 6$ s.

c. $t^2 - 8t - 20 = 0$: $t = 10$ s; reject -2.

d

d. Check endpoints and v's turning point:

$v(0) = 24$,

$v(4) = -8$,

$v(5) = -6$.

Greatest speed $= 24$ m/s.

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Solution 3

6. A particle starts at the origin with velocity $v = 8t - 2t^2$ m/s, $t \geq 0$.

a. When is it next instantaneously at rest?

b. Find its greatest speed on $0 \leq t \leq 4$.

a

a. $2t(4 - t) = 0$. The next time is $t = 4$ s.

b

b. Velocity is non-negative throughout the interval.

$$\frac{dv}{dt} = 8 - 4t = 0 \Rightarrow t = 2.$$

$v(2) = 8$ m/s; both endpoint velocities are zero. Greatest speed: 8 m/s.

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Solution 4

7. A particle leaves O and later returns to O. Its distance from O is modelled by

$$s = 3t^2 - t^3, \quad 0 \leq t \leq T.$$

The model is valid while $s \geq 0$. Find T and explain.

7

$s = t^2(3 - t)$.

The non-zero root is $t = 3$. For $t > 3$, the expression becomes negative and cannot represent distance.

7

The particle returns to O at $T = 3$ s. The stated distance model ends there.

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Solution 5

8. A particle has velocity
$v = \frac{1}{5}(3t^2 - 10t + 3), \quad t \geq 0.$

a. Find the times when it is at rest.

b. Find its greatest speed on $0 \leq t \leq 3$.

a

a. $(3t - 1)(t - 3) = 0.$

$t = \frac{1}{3}$ s or 3 s.

b

b. $a = (6t - 10)/5 = 0$ at $t = 5/3.$

$v(0) = \frac{3}{5},$

$v(5/3) = -\frac{16}{15},$

$v(3) = 0.$

Greatest speed $= 16/15$ m/s.

Using Differentiation

Solution 6

Using differentiation

$$v = \frac{dx}{dt}, \qquad a = \frac{dv}{dt} = \frac{d^2x}{dt^2}.$$

A particle has $x = t^4 - 32t + 14$ m. Find:

a. velocity at $t = 3$;

b. time at rest;

c. acceleration at $t = 1.5$.

a

a. $v = 4t^3 - 32$, so $v(3) = 76$ m/s.

b

b. $4t^3 - 32 = 0 \Rightarrow t = 2$ s.

c

c. $a = 12t^2$, so $a(1.5) = 27$ m/s^2.

Test Your Understanding

Solution 7

A cat's displacement from a house is $s = t^3 - \frac{3}{2}t^2 - 36t$ m, $\quad t \geq 0$.

a. Find its velocity at $t = 2$.

b. When is it at rest?

c. Find its acceleration at $t = 5$.

a

a. $v = 3t^2 - 3t - 36$.

$v(2) = -30$ m/s.

b

b. $(t + 3)(t - 4) = 0$. Reject negative time: $t = 4$ s.

c

c. $a = 6t - 3$, so $a(5) = 27$ m/s^2.

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Solution 8

5. A particle has velocity $v = kt - 3t^2$ m/s. Its initial acceleration is 4 m/s^2.

a. Find k.

b. Find its acceleration when it is next instantaneously at rest.

a

a. $a = k - 6t$. At $t = 0$, $a = 4$, hence $k = 4$.

b

b. $v = t(4 - 3t) = 0$: next rest at $t = 4/3$ s.

$a = 4 - 6(4/3) = -4$ m/s^2.

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Solution 9

6. A printer head has displacement $s = \frac{1}{4}(4t^3 - 15t^2 + 12t + 30)$ cm, $0 \le t \le 3$.

Find the distance between the points at which it is instantaneously at rest, to 1 d.p.

6

$v = \frac{1}{4}(12t^2 - 30t + 12) = 0$

$2t^2 - 5t + 2 = (2t - 1)(t - 2) = 0.$

Rest at $t = 0.5$ s and $t = 2$ s.

6

$s(0.5) = 8.1875, \qquad s(2) = 6.5.$

Separation $= 8.1875 - 6.5 = 1.6875$ cm.

Answer: 1.7 cm (1 d.p.).

Maxima and Minima Problems

Solution 10

A yo-yo's distance below the hand is $s = 0.6t + 0.4t^2 - 0.2t^3, \quad 0 \le t \le 3.$

a. Justify the time restriction.

b. Find the maximum distance from the hand to 3 s.f.

a

a. $s = \frac{1}{5}t(3 - t)(1 + t)$. It is non-negative for $0 \le t \le 3$, returning to the hand at $t = 3$.

b

b. $v = 0.6 + 0.8t - 0.6t^2 = 0.$

$$t = \frac{2 + \sqrt{13}}{3} = 1.8685\ldots$$

The other root is negative. Substitution gives maximum distance 1.21 m.

Test Your Understanding

Solution 11

1. A dolphin's velocity is $v = t^3 - 16t^2 + 64t$ m/s until the first maximum, after which it maintains that velocity. Find the time and maximum velocity.
2. A particle has $v = 2t^2 - 14t + 20$, $t \geq 0$. Find its times at rest and greatest speed on $0 \leq t \leq 4$.

1

1. $a = 3t^2 - 32t + 64$.

$a = (3t - 8)(t - 8)$.

The first maximum is at $t = 8/3$ s.

$v(8/3) = 2048/27 = 75.9$ m/s.

2

2. $v = 2(t - 2)(t - 5)$: rest at 2 s and 5 s.

Check $v(0) = 20$, $v(3.5) = -4.5$, $v(4) = -4$. Greatest speed: 20 m/s.

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Solution 12

5. A particle starts at O; its distance from O is $s = \frac{9}{2}t^2 - t^3, \quad 0 \leq t \leq 4.5$.

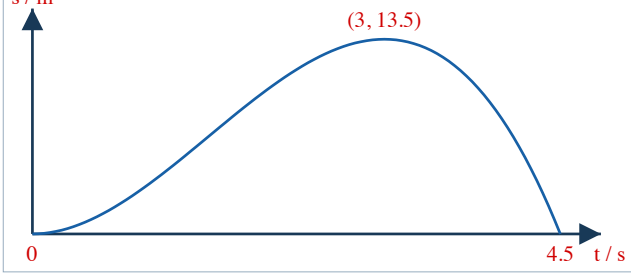
a. Sketch its displacement–time graph.

b. Justify the restriction.

c. Find its greatest distance from O.

d. Find the acceleration magnitude there.

a · Sketch



a–b

a–b. $s = t^2(4.5 - t)$: starts with a horizontal tangent at $(0, 0)$, rises to a maximum, then returns to $(4.5, 0)$. Beyond 4.5 s it is negative, so cannot represent distance.

c

c. $v = 9t - 3t^2 = 3t(3 - t)$.

The maximum occurs at $t = 3$: $s = 13.5$ m.

d

d. $a = 9 - 6t$. At $t = 3$, $a = -9$ m/s², magnitude 9 m/s².

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Solution 13

6. A train moves along a straight 4 km test track. Its displacement from the start is modelled by

$s = 3.6t + 1.76t^2 - 0.02t^3.$

$0 \le t \le 90.$

Show that it never reaches the far end of the track.

6

$v = 3.6 + 3.52t - 0.06t^2 = 0$

$3t^2 - 176t - 180 = 0.$

The roots are approximately -1.005 and 59.672 s.

6

Velocity changes from positive to negative at 59.672 s, giving the maximum.

$s_{\max} = 2230$ m (3 s.f.) < 4000 m.

The train never reaches the end.

Negative later displacement means it has passed the starting point in the opposite direction.

Using Integration

Solution 14

Using integration

$v = \int a\,dt, \qquad x = \int v\,dt.$

Use initial conditions to find each constant.

A particle has $v = 6t - t^2$, $t \ge 0$, and $x(0) = 5$.

a. Find $x(t)$.

b. Find its distance from its starting position at $t = 6$.

a

a. $x = 3t^2 - \frac{1}{3}t^3 + C.$

At $t = 0$, $x = 5$, so $C = 5$.

$x = 3t^2 - \frac{1}{3}t^3 + 5.$

b

b. $x(6) = 41$, so distance from starting position is $|41 - 5| = 36$ m.

Further Example

A particle has velocity $v = 5 - 3t^2$ m/s, $t \geq 0$. Find the distance travelled in the third second of its motion.

Solution 15

The third second is $2 \leq t \leq 3$. Velocity is negative throughout this interval, so distance is the negative of displacement.

$$\Delta s = \int_2^3 (5 - 3t^2)\,dt = [5t - t^3]_2^3$$

$$= -12 - 2 = -14 \text{ m}.$$

Distance travelled: 14 m.

Further Example

A particle has $v = 2t^2 - 9t + 4$ m/s and $x(0) = 15$ m.

a. Find the times at rest.

b. Find acceleration at $t = 5$.

c. Find total distance travelled on $0 \leq t \leq 5$.

Solution 16

a

a. $(2t - 1)(t - 4) = 0$: $t = \frac{1}{2}, 4$ s.

b. $a = 4t - 9$, so $a(5) = 11$ m/s^2.

c

c. Integrate:

$$x = \tfrac{2}{3}t^3 - \tfrac{9}{2}t^2 + 4t + 15.$$

Velocity changes sign at both rest times. Add the three positive distances:

$$\tfrac{23}{24} + \tfrac{343}{24} + \tfrac{25}{6} = \tfrac{233}{12} \text{ m}.$$

8. A particle starts at O with velocity $v = 6t^2 - 26t + 15$ m/s.

Find the two non-zero times when it passes through O.

8

$x = \int v\,dt = 2t^3 - 13t^2 + 15t + C.$

Since $x(0) = 0$, $C = 0$.

8

$x = t(2t - 3)(t - 5) = 0.$

Excluding $t = 0$, it passes through O at 1.5 s and 5 s.

9. A particle starts at O with velocity $v = 3t^2 - 12t + 5$ m/s.

a. When does it return to O?

b. Find the distance travelled on $2 \le t \le 3$.

a

a. $x = t^3 - 6t^2 + 5t + C$; $C = 0$.

$x = t(t - 1)(t - 5) = 0.$

Returns at $t = 1, 5$ s.

b

b. Velocity is negative throughout $[2, 3]$; its zeros are approximately 0.473 and 3.528.

$x(2) = -6, \qquad x(3) = -12.$

Distance: $|-12 - (-6)| = 6$ m.

10. A particle has acceleration $a = 4t - 3$ m/s^2 and initial velocity 4 m/s. At $t = T > 0$, its velocity is again 4 m/s. Find T.

10

$v = 2t^2 - 3t + C.$

Initial velocity gives $C = 4.$

10

$4 = 2T^2 - 3T + 4 \Rightarrow T(2T - 3) = 0.$

Since $T > 0$, $T = 1.5$ s.

Constant acceleration formulae

A particle has constant acceleration a, initial velocity u, and initial displacement zero. Use integration to prove:

a. $v = u + at$;

b. $s = ut + \frac{1}{2}at^2$.

a

a. $v = \int a\,dt = at + C.$

At $t = 0$, $v = u$, so $C = u.$

$v = u + at.$

b

b. $s = \int (u + at)\,dt = ut + \frac{1}{2}at^2 + K.$

At $t = 0$, $s = 0$, so $K = 0.$

$s = ut + \frac{1}{2}at^2.$

3. Prove that $s = ut + \frac{1}{2}at^2$, with u, a constant, describes constant acceleration a.
4. Which displacement functions have constant acceleration?

A. $s = 2t^2 - t^3$

B. $s = 4t + 7$

C. $s = t^2/4$

D. $s = 3t - 2/t^2$

E. $s = 6$

3. $v = u + at$, then $dv/dt = a$, a constant.

4. Differentiate twice:

A. $4 - 6t$: not constant.

B. 0: constant (uniform velocity).

C. $1/2$: constant.

D. $-12/t^4$: not constant.

E. 0: constant (stationary).

6. A train passes A at $t = 0$ and B at $t = 40$. Its distance from A is $s = 25t - 0.2t^2, \quad 0 \le t \le 40$.
   a. Find AB.
   b. Show acceleration is constant.
   c. At $t = 0$, a bird passes B towards A at 7 m/s, with constant acceleration 0.6 m/s^2 towards A. How far from A is it when directly above the train?

a

a. $AB = s(40) = 1000 - 320 = 680$ m.

b. $v = 25 - 0.4t$, so $a = -0.4$ m/s^2, constant.

c

c. Bird's position from A: $680 - 7t - 0.3t^2$.

Equate positions:

$25t - 0.2t^2 = 680 - 7t - 0.3t^2$

$t^2 + 320t - 6800 = (t - 20)(t + 340) = 0.$

$t = 20$ s; distance $s(20) = 420$ m.